

Vitamins

Vitamins

- ▶ are a group of organic nutrients, required in small quantities for a variety of biochemical functions that, generally, cannot be synthesized by the body and must therefore be supplied in the diet.
- ▶ an organic compound that is required in the diet in small amounts for the maintenance of normal metabolic integrity.

Vitamins

➤ Classified according to solubility

1- Vitamins that function in the watery areas
= water-soluble vitamins

2- Vitamins that function in fatty tissues =
fat-soluble vitamins

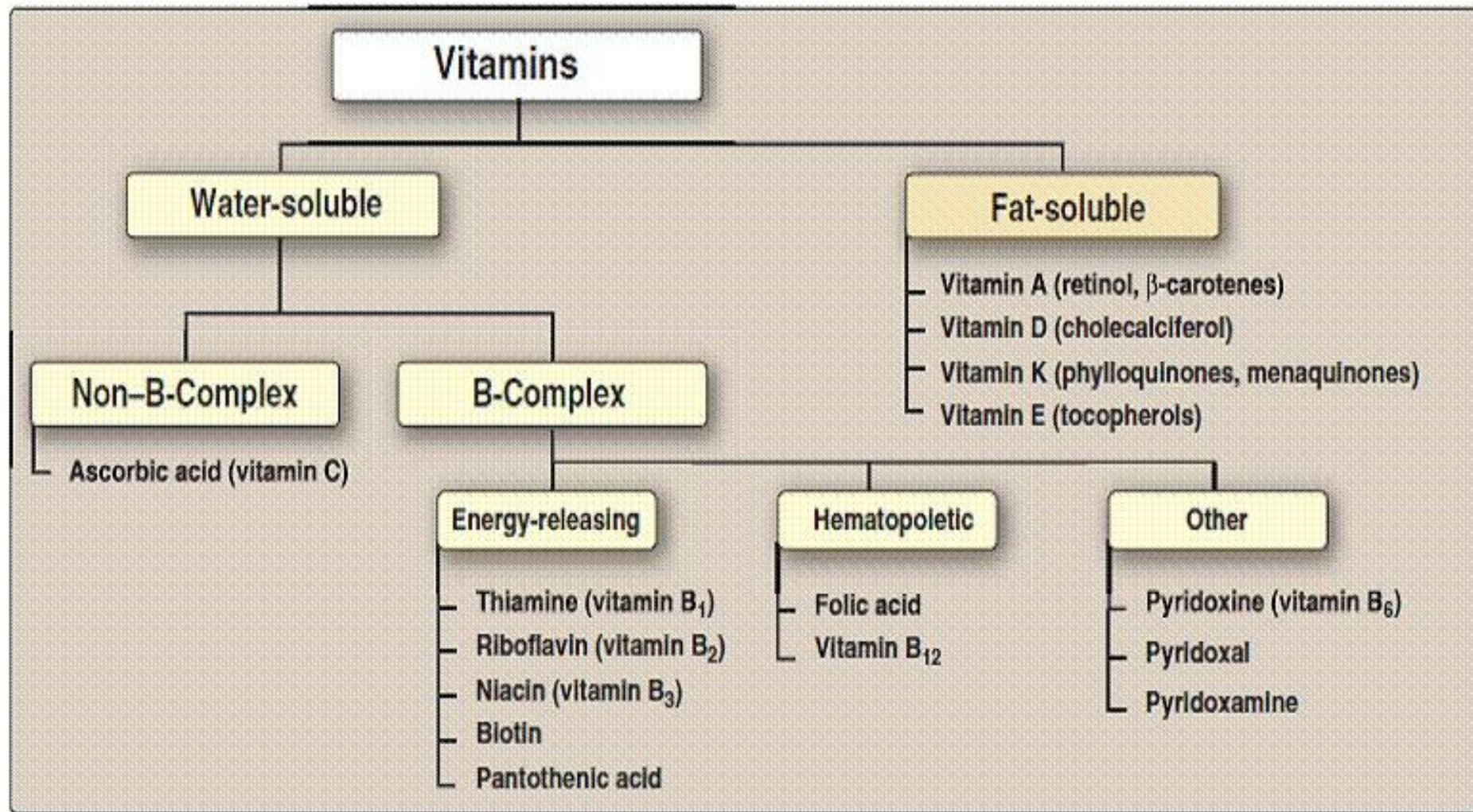
lipid-soluble vitamins

- ▶ absorbed only when there is normal fat absorption.
- ▶ are transported in the blood in lipoproteins or attached to specific binding proteins.
- ▶ dietary inadequacy, conditions affecting the digestion and absorption of the lipid soluble vitamins can all lead to deficiency syndromes
- ▶ Toxicity can result from excessive intake of vitamins A and D.
- ▶ Vitamin A and the carotenes and vitamin E are antioxidants and have possible roles in prevention of atherosclerosis and cancer

Water-soluble vitamins

- ▶ are vitamins B and C, folic acid, biotin and pantothenic acid
- ▶ they function mainly as enzyme cofactors.
- ▶ Deficiency of a single vitamin of the B complex is rare since poor diets are most often associated with **multiple deficiency states**. Nevertheless, specific syndromes are characteristic of deficiencies of individual vitamins

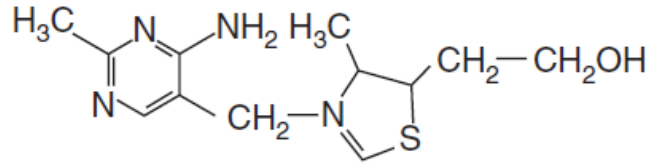
Vitamin		Functions	Deficiency Disease
Lipid-soluble			
A	Retinol, β -carotene	Visual pigments in the retina; regulation of gene expression and cell differentiation (β -carotene is an antioxidant)	Night blindness, xerophthalmia; keratinization of skin
D	Calciferol	Maintenance of calcium balance; enhances intestinal absorption of Ca^{2+} and mobilizes bone mineral; regulation of gene expression and cell differentiation	Rickets = poor mineralization of bone; osteomalacia = bone demineralization
E	Tocopherols, tocotrienols	Antioxidant, especially in cell membranes; roles in cell signaling	Extremely rare—serious neurologic dysfunction
K	Phylloquinone: menaquinones	Coenzyme in formation of γ -carboxyglutamate in enzymes of blood clotting and bone matrix	Impaired blood clotting, hemorrhagic disease
Water-soluble			
B ₁	Thiamin	Coenzyme in pyruvate and α -ketoglutarate dehydrogenases, and transketolase; regulates Cl^- channel in nerve conduction	Peripheral nerve damage (beriberi) or central nervous system lesions (Wernicke-Korsakoff syndrome)
B ₂	Riboflavin	Coenzyme in oxidation and reduction reactions (FAD and FMN); prosthetic group of flavoproteins	Lesions of corner of mouth, lips, and tongue, seborrheic dermatitis
Niacin	Nicotinic acid, nicotinamide	Coenzyme in oxidation and reduction reactions, functional part of NAD and NADP; role in intracellular calcium regulation and cell signaling	Pellagra—photosensitive dermatitis, depressive psychosis
B ₆	Pyridoxine, pyridoxal, pyridoxamine	Coenzyme in transamination and decarboxylation of amino acids and glycogen phosphorylase; modulation of steroid hormone action	Disorders of amino acid metabolism, convulsions
	Folic acid	Coenzyme in transfer of one-carbon fragments	Megaloblastic anemia
B ₁₂	Cobalamin	Coenzyme in transfer of one-carbon fragments and metabolism of folic acid	Pernicious anemia = megaloblastic anemia with degeneration of the spinal cord
	Pantothenic acid	Functional part of CoA and acyl carrier protein: fatty acid synthesis and metabolism	Peripheral nerve damage (nutritional melalgia or “burning foot syndrome”)
H	Biotin	Coenzyme in carboxylation reactions in gluconeogenesis and fatty acid synthesis; role in regulation of cell cycle	Impaired fat and carbohydrate metabolism, dermatitis
C	Ascorbic acid	Coenzyme in hydroxylation of proline and lysine in collagen synthesis; antioxidant; enhances absorption of iron	Scurvy—impaired wound healing, loss of dental cement, subcutaneous hemorrhage



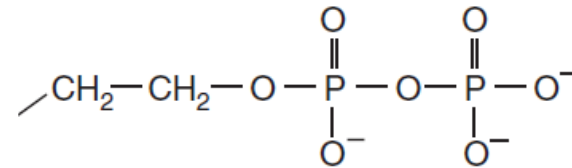
VITAMIN B1 (THIAMIN)

- ▶ Found in a wide variety of foods, including meat, grains, white bread, cereal.
- ▶ Has a role in energy-yielding metabolism, and especially the metabolism of carbohydrates
- ▶ **Thiamin diphosphate** is the coenzyme for transketolase and multienzyme complexes that catalyze oxidative decarboxylation reactions “pyruvate dehydrogenase ; α -ketoglutarate dehydrogenase” ; and the branched-chain keto-acid dehydrogenase
- ▶ has a role in nerve conduction

VITAMIN B1 (THIAMIN)



Thiamin



Thiamin diphosphate

VITAMIN B1 (THIAMIN) deficiency

- ▶ A chronic peripheral neuritis, **beriberi**, which may or may not be associated with **heart failure** and **edema**;
- ▶ acute pernicious (fulminating) beriberi (shoshin beriberi), in which heart failure and metabolic abnormalities predominate, without peripheral neuritis; and
- ▶ **Wernicke encephalopathy** with **Korsakoff psychosis**, which is associated especially with alcohol and narcotic abuse.
- ▶ **Can Be Assessed by Erythrocyte Transketolase Activation**

Beriberi

- ▶ Weakness, nerve degeneration, irritability, poor arm/leg coordination, loss of nerve transmission(Dry beriberi)
- ▶ Edema, enlarged heart, heart failure(wet beriberi)
- ▶ Symptoms due to poor metabolism of glucose
- ▶ Depression and weakness can be seen after only 10 days on a thiamin-free diet

THIAMINE DEFICIENCY (Beriberi)

Common early manifestations



Foot Drop

Dyspnea, orthopnea



**WET
BERIBERI**

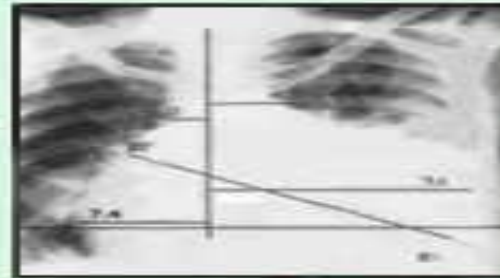
DRY BERIBERI

Emaciation

Great Weakness

Wrist Drop

Aphonia may appear
(Poor prognosis;
vagus nerve involved)



Dilatation of
right heart;
heart failure

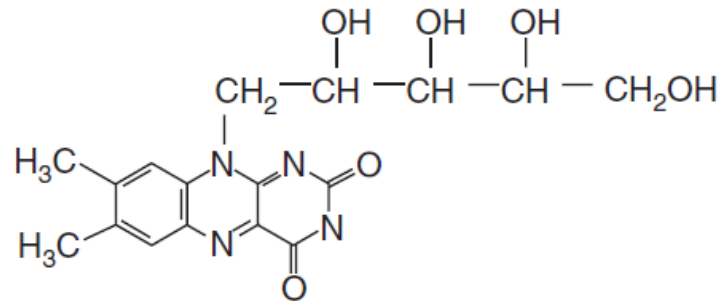
Wernicke's
Syndrome

Ophthalmoplegia
→ Confusion
→ Coma
→ Death



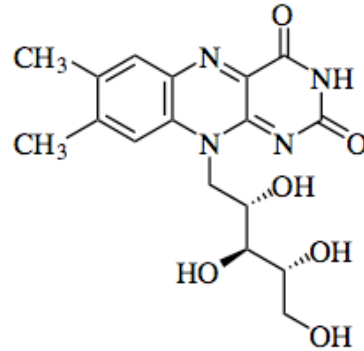
VITAMIN B2 (RIBOFLAVIN)

- ▶ Sources: milk ,egg ,green leafy vegetables
- ▶ **ROLE IN ENERGY YIELDING METABOLISM**
- ▶ flavin mononucleotide (FMN) and flavin adenine dinucleotide (FAD) are Electron Carriers in Oxidoreduction Reactions
- ▶ mitochondrial respiratory chain, key enzymes in fatty acid and amino acid oxidation, purine degradation and the citric acid cycle.



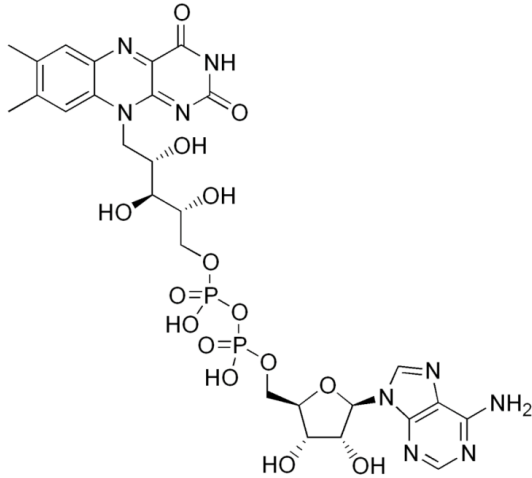
Riboflavin

Vitamin B2 - riboflavin



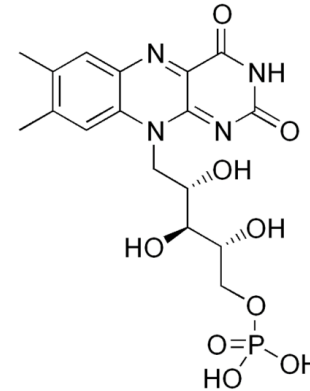
Riboflavin is a precursor for FAD and FMN.

FAD - flavin adenine dinucleotide.



FAD is a cofactor for pyruvate dehydrogenase complex (PDC), and succinate dehydrogenase in TCA cycle.

FMN - Flavin mononucleotide



FMN is an electron carrier in the electron transport chain.

VITAMIN B2 (RIBOFLAVIN)

- ▶ deficiency occurs in many countries, it is not fatal, because there is very efficient conservation of tissue riboflavin.
- ▶ Deficiency is characterized by cheilosis, desquamation and inflammation of the tongue, and a seborrheic dermatitis.
- ▶ Riboflavin nutritional status is assessed by measurement of the activation of erythrocyte glutathione reductase by FAD added in vitro, and urinary flavin excretion.





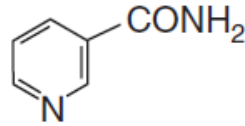
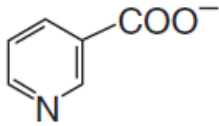


VITAMIN B3 (NIACIN)

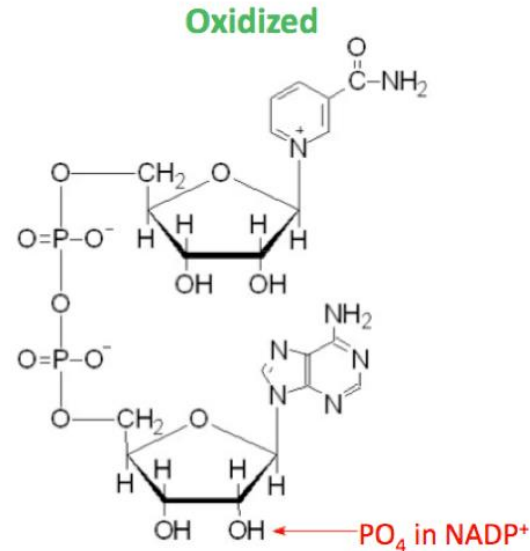
- ▶ can be synthesized in the body from tryptophan, refined cereal, milk, liver, enriched grains, beef, chicken, turkey, fish and peanuts
- ▶ **nicotinic acid** and **nicotinamide**, “coenzymes **NAD** and **NADP**” in oxidation/reduction reactions,
- ▶ is the source of ADP-ribose for the **ADP-ribosylation** of proteins and nucleoproteins involved in the **DNA repair mechanism**
- ▶ Cyclic ADP-ribose and nicotinic acid adenine dinucleotide increase intracellular calcium in response to neurotransmitters and hormones
- ▶ Daily urinary excretion of methylated metabolites
E.g. The ratio of 2-pyridone to *N*¹-methyl-nicotinamide in urine.

VITAMIN B3 (NIACIN)

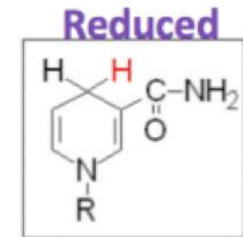
NAD^+ A cofactor to remember.



Niacin (nicotinic acid and nicotinamide)



- Made up of nicotinamide attached to an AMP
Can accept two electrons when reduced
Gets reduced to NADH



VITAMIN B3 (NIACIN) deficiency 3/4D's

- ▶ Pellagra is characterized by a photosensitive dermatitis, dementia and diarrhea. Untreated pellagra is fatal.
- ▶ Although the nutritional etiology of pellagra is well established, and either tryptophan or niacin prevents or cures the disease, additional factors, including deficiency of riboflavin or vitamin B6, both of which are required for synthesis of nicotinamide from tryptophan, may be important.

Pellagra

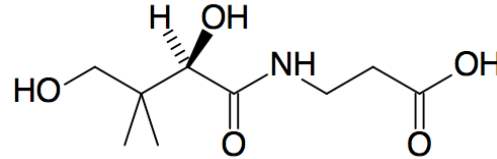


PANTOTHENIC ACID

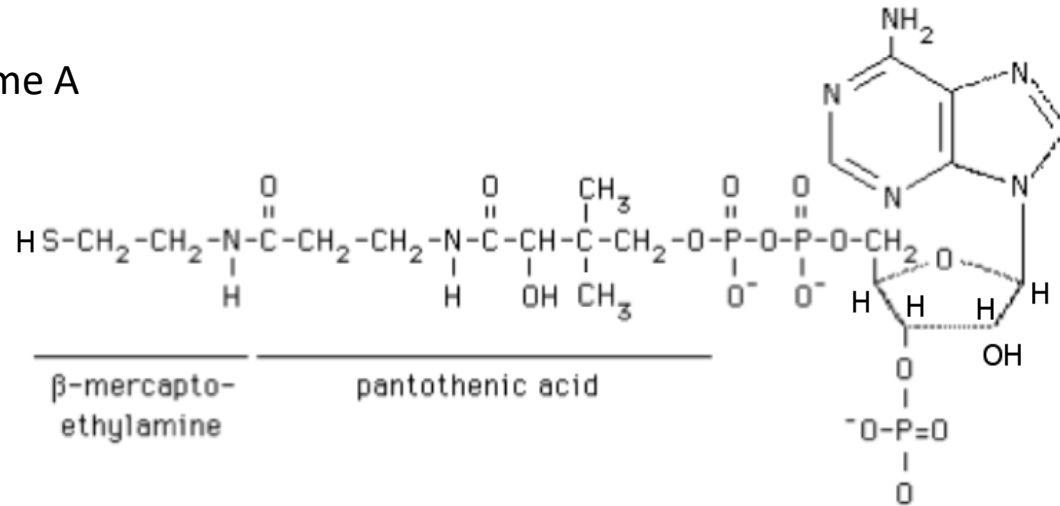
- ▶ in all food-stuffs
- ▶ has a central role in acyl group metabolism
- ▶ when acting as the pantetheine functional moiety of coenzyme A or acyl carrier protein (ACP), and acetylation reaction of drugs

Vitamin B5 - pantothenic acid (needed for making CoA)

pantothenic acid



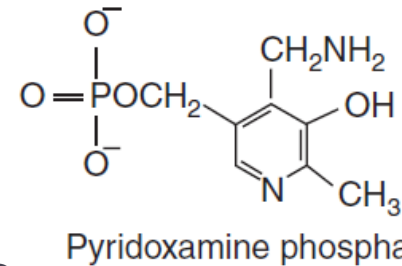
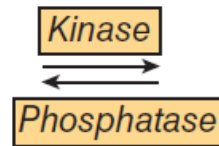
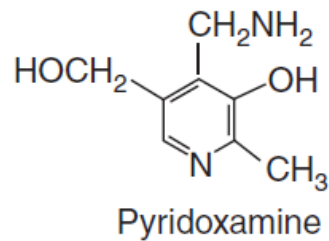
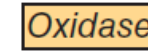
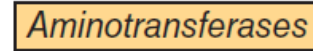
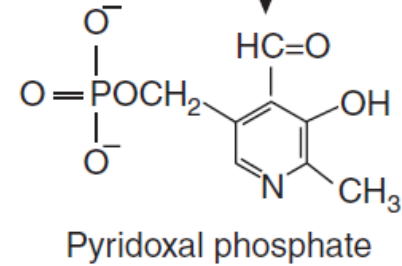
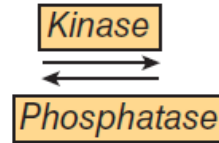
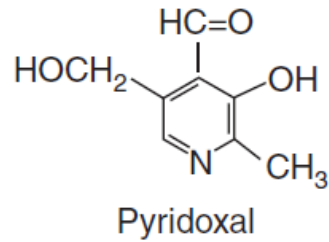
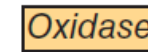
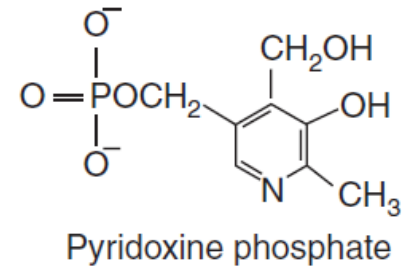
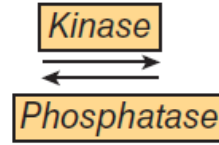
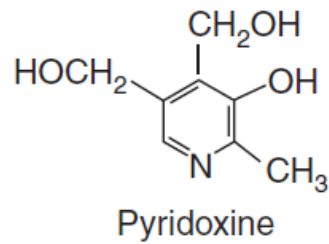
Coenzyme A



We get pantothenic acid in our diet as CoA, which must be broken down to pantothenic acid to be absorbed in intestine. We then use the pantothenic acid in making our own CoA.

VITAMIN B6

- ▶ pyridoxine, pyridoxal, pyridoxamine
- ▶ wheat ,egg, liver, corn
- ▶ **AMINO ACID & GLYCOGEN METABOLISM & IN STEROID HORMONE ACTION**
- ▶ Aminoacid transamination and decarboxylation, glycogen phosphorylase, heme biosynthesis (δ -Aminolevulinate synthase)
- ▶ The most widely used method of assessing vitamin B6 status is by the activation of erythrocyte transaminases by pyridoxal phosphate added in vitro, expressed as the activation coefficient.



VITAMIN B6

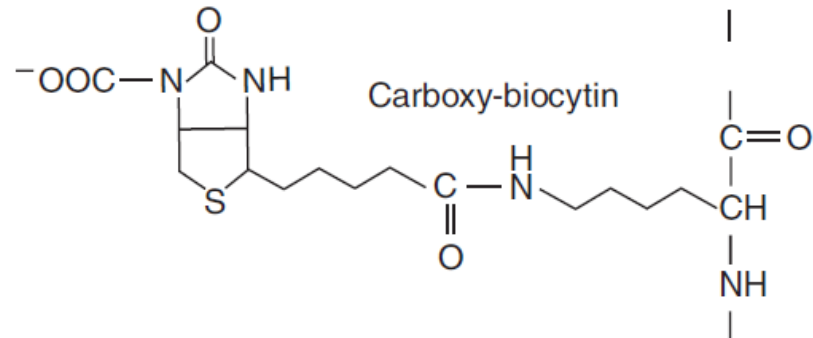
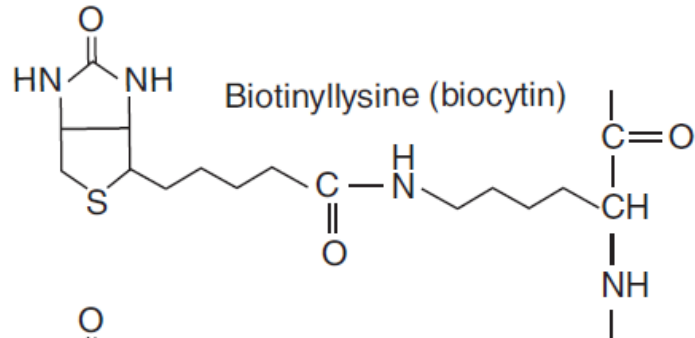
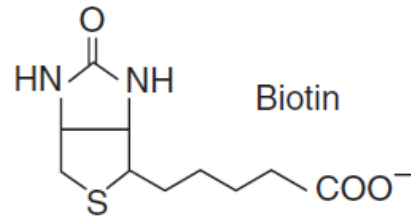
- ▶ clinical deficiency disease is rare, there is evidence that a significant proportion of the population have marginal vitamin B6 status. Moderate deficiency results in abnormalities of tryptophan and methionine metabolism. Increased sensitivity to steroid hormone action may be important in the development of **hormone-dependent cancer** of the breast, uterus, and prostate, and vitamin B6 status may affect the prognosis.
- ▶ isoniazid therapy (Tuberculosis treatment) cause deficiency

VITAMIN B6 deficiency

- Sideroblastic anemia (Iron deficiency anemia, The body has iron available, but cannot incorporate it into hemoglobin. Sideroblasts are seen, which are nucleated erythrocytes with granules of iron in their cytoplasm)
- Cheilosis & stomatitis (cracking of the lips and of the corners of the mouths respectively).aa
- Convulsions

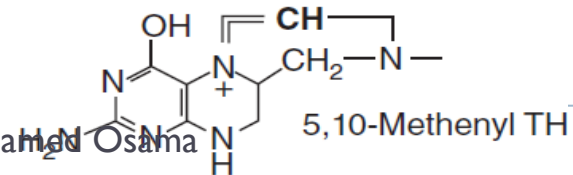
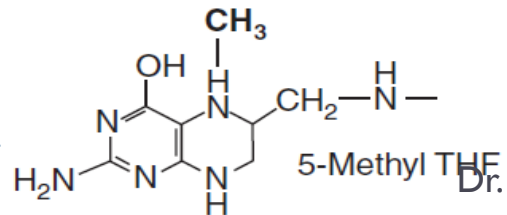
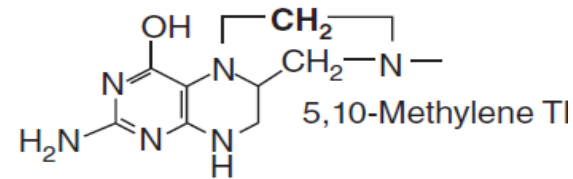
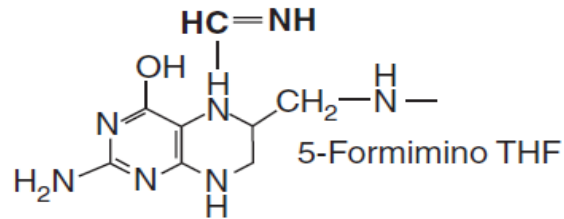
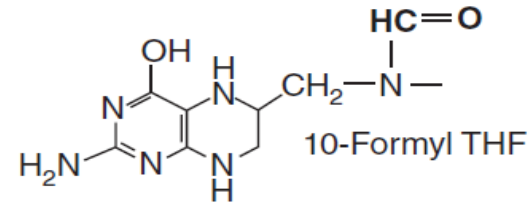
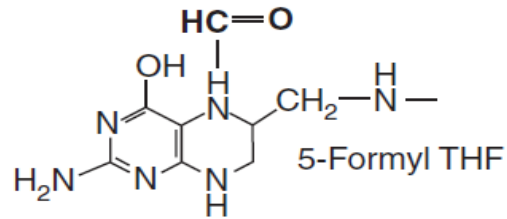
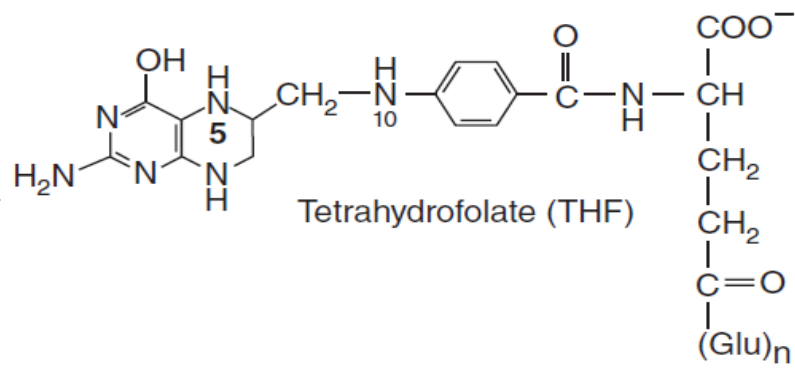
BIOTIN

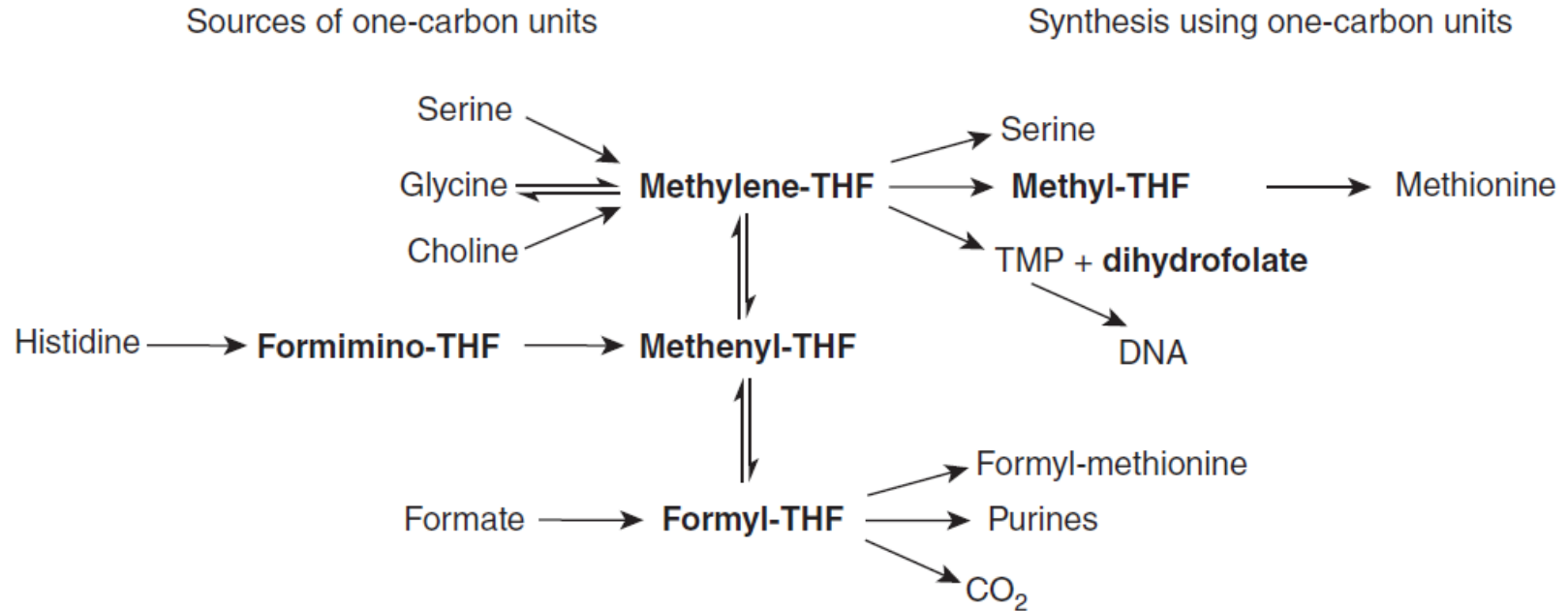
- ▶ source:milk,egg yolk,liver
- ▶ **a Coenzyme of Carboxylase Enzymes** acetyl-CoA carboxylase, pyruvate carboxylase, propionyl-CoA carboxylase, and methylcrotonyl-CoA carboxylase
- ▶ Deficiency is unknown, except among people maintained for many months on total parenteral nutrition, and a very small number who eat abnormally large amounts of uncooked egg white, which contains avidin, a protein that binds biotin and renders it unavailable for absorption.
- ▶ Deficiency leads to Alopecia (hair loss), bowel inflammation, muscle pain

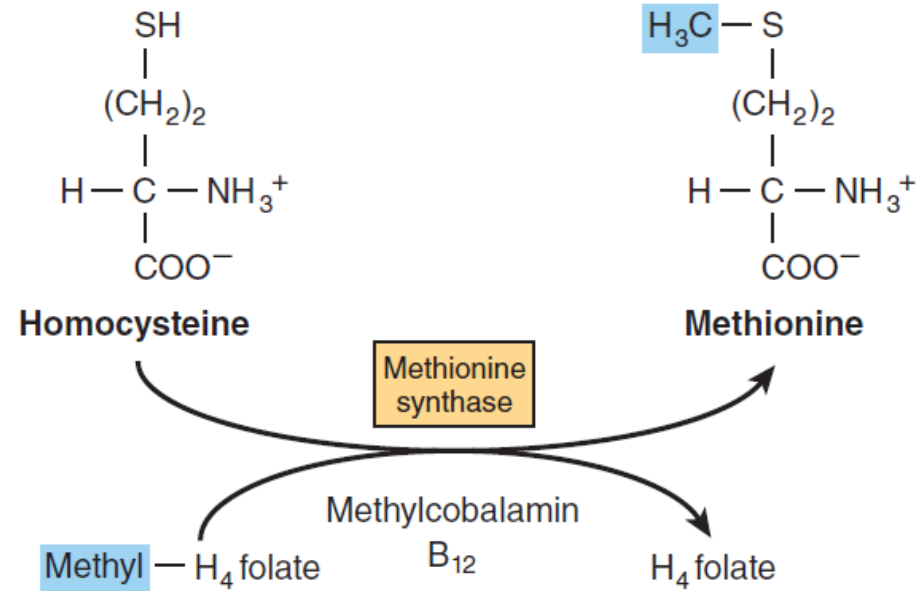


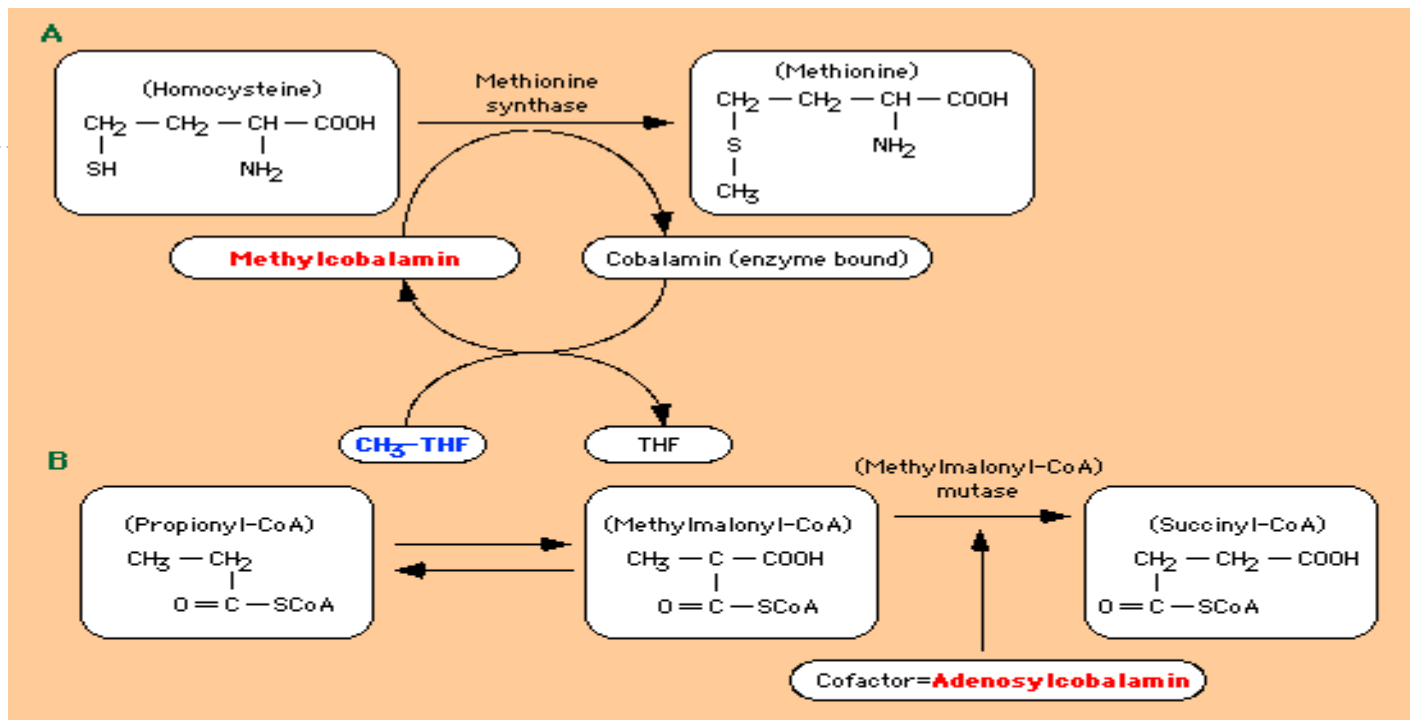
FOLIC ACID

- ▶ yeast, leafy vegetable
- ▶ Tetrahydrofolate
- ▶ Carrier of One-Carbon Units serine hydroxymethyl transferase, thymidylate synthase, homocysteine metabolism (homocysteine methyl transferase)
- ▶ Methotrexate, an analog of 10-methyl-tetrahydrofolate, inhibits dihydrofolate reductase and has been exploited as an anticancer drug
- ▶ trimethoprim and antimalarial drugs inhibit dihydrofolate reductase



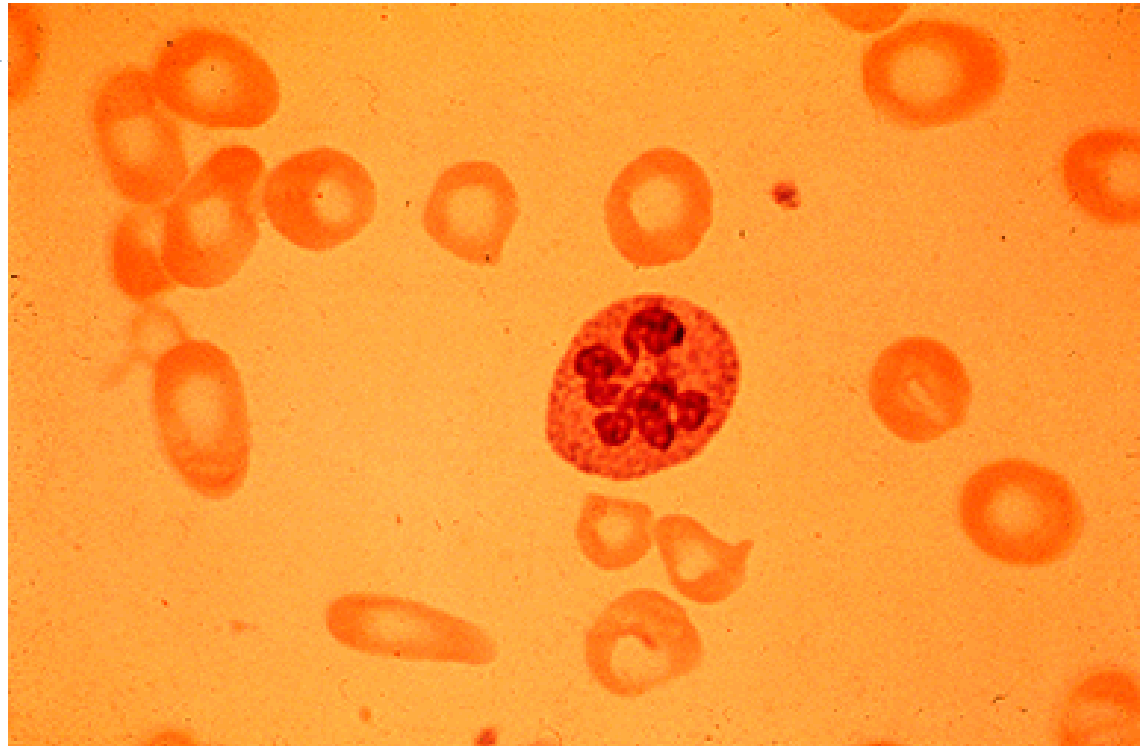






Actions of cobalamin Role of cobalamin in homocysteine and methylmalonic acid metabolism. Panel A – Methylcobalamin is a cofactor in the synthesis of methionine from homocysteine. Panel B – Adenosylcobalamin is a cofactor in the synthesis of succinyl-CoA from methylmalonyl-CoA. Tetrahydrofolate (THF) participates in homocysteine but not methylmalonic acid (MMA) metabolism. Thus, cobalamin deficiency is characterized by elevations in the serum levels of both homocysteine and MMA, while only homocysteine levels are elevated in folate deficiency. (Reproduced with permission from Tefferi, A, Pruthi, RK. The Biochemical Basis of Cobalamin Deficiency. Mayo Clin Proc 1994; 69:181).

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- ▶ Deficiency of folic acid itself or deficiency of vitamin B12, which leads to functional folic acid deficiency, affects cells that are dividing rapidly because they have a large requirement for thymidine for DNA synthesis. Clinically, this affects the bone marrow, leading to megaloblastic anemia.

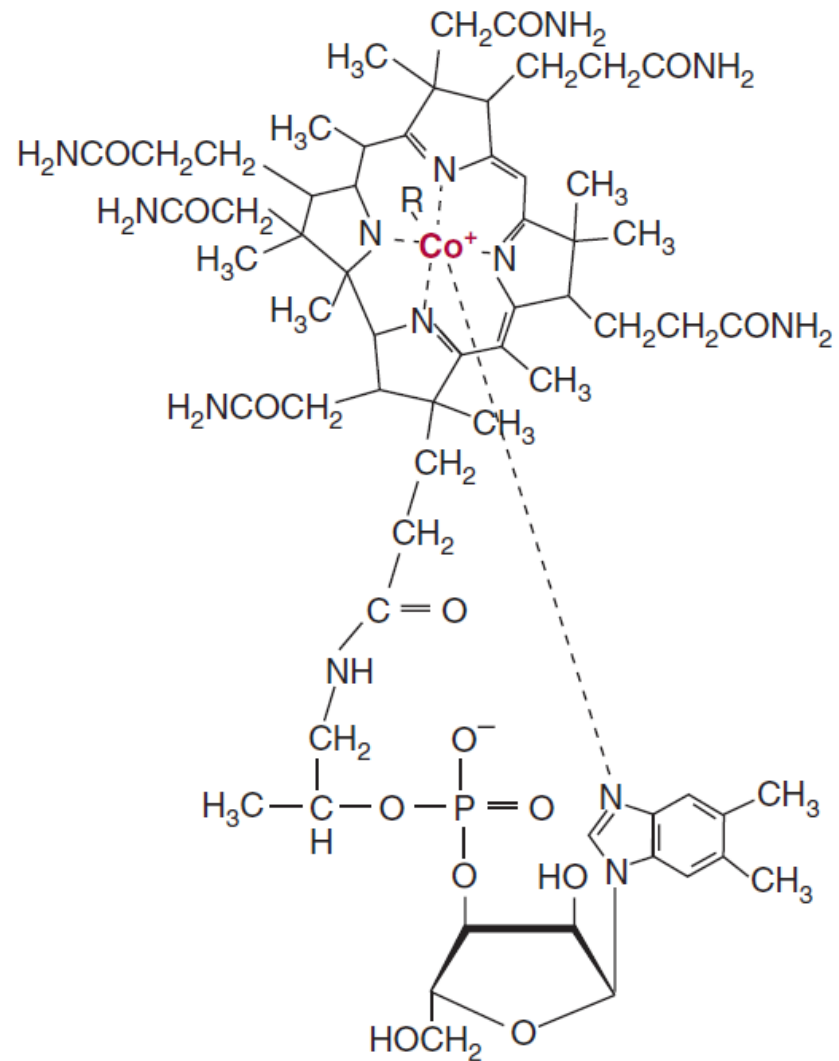


Megaloblastic blood picture Peripheral blood smear showing a hypersegmented neutrophil (7 lobes) and macroovalocytes, a pattern that can be seen with cobalamin or folate deficiency. Courtesy of Stanley L Schrier, MD.

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- ▶ **Folic Acid Supplements Reduce the Risk of Neural Tube Defects & Hyperhomocysteinemia, & May Reduce the Incidence of Cardiovascular Disease & Some Cancers**
 - ▶ Folic acid supplements will rectify the megaloblastic anemia of vitamin B12 deficiency but not the irreversible nerve damage seen in vitamin B12 deficiency

VITAMIN B12

- ▶ **Cobalamins**
- ▶ is found only in foods of animal origin
- ▶ **Methylmalonyl CoA mutase**, and **methionine synthase**
- ▶ accumulation of methylmalonyl CoA and urinary excretion of methylmalonic acid, provides means of assessing vitamin B12 nutritional status



VITAMIN B12

- ▶ **intrinsic factor**, a small glycoprotein secreted by the parietal cells of the gastric mucosa.
- ▶ is absorbed bound to intrinsic factor.
- ▶ absorbed from the distal third of the ileum via receptors that bind the intrinsic factor-vitamin B12 complex, but not free intrinsic factor or free vitamin.
- ▶ There is considerable enterohepatic circulation of vitamin B12, with excretion in the bile, then reabsorption after binding to intrinsic factor in the ileum.

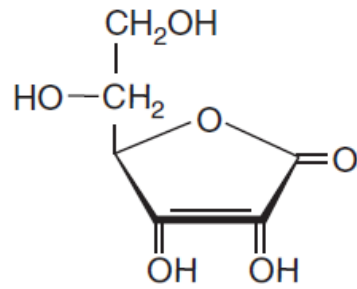
VITAMIN B12 deficiency

- ▶ Pernicious anemia arises when vitamin B12 deficiency impairs the metabolism of folic acid, leading to functional folate deficiency that disturbs erythropoiesis, causing immature precursors of erythrocytes to be released into the circulation (megaloblastic anemia)
- ▶ There is irreversible degeneration of the spinal cord in pernicious anemia, as a result of failure of methylation of one arginine residue in myelin basic protein. This is the result of methionine deficiency in the central nervous system, rather than secondary folate deficiency.

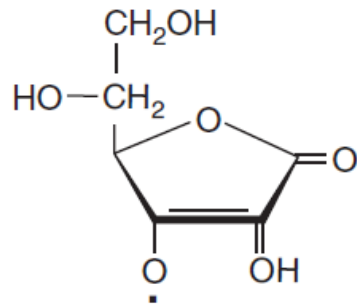
Vitamin	Coenzyme	Typical reaction type	Consequences of deficiency
Thiamine (B ₁)	Thiamine pyrophosphate	Aldehyde transfer	Beriberi (weight loss, heart problems, neurological dysfunction)
Riboflavin (B ₂)	Flavin adenine dinucleotide (FAD)	Oxidation–reduction	Cheliosis and angular stomatitis (lesions of the mouth), dermatitis
Pyridoxine (B ₆)	Pyridoxal phosphate	Group transfer to or from amino acids	Depression, confusion, convulsions
Nicotinic acid (niacin)	Nicotinamide adenine dinucleotide (NAD ⁺)	Oxidation–reduction	Pellagra (dermatitis, depression, diarrhea)
Pantothenic acid	Coenzyme A	Acyl-group transfer	Hypertension
Biotin	Biotin–lysine adducts (biocytin)	ATP-dependent carboxylation and carboxyl-group transfer	Rash about the eyebrows, muscle pain, fatigue (rare)
Folic acid	Tetrahydrofolate	Transfer of one-carbon components; thymine synthesis	Anemia, neural-tube defects in development
B ₁₂	5'-Deoxyadenosyl cobalamin	Transfer of methyl groups; intramolecular rearrangements	Anemia, pernicious anemia, methylmalonic acidosis

ASCORBIC ACID Vitamin C

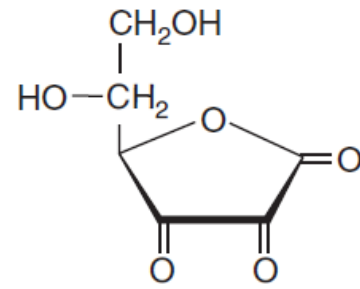
- ▶ roles in the copper-containing hydroxylases and the α -ketoglutarate-linked iron-containing hydroxylases.
- ▶ nonenzymic effects as a result of its action as a reducing agent and oxygen radical quencher
- ▶ **peptidylglycine hydroxylase, Dopamine β -hydroxylase, Proline and lysine hydroxylases**
- ▶ enhances the absorption of inorganic iron
- ▶ high doses of vitamin C prevent the common cold, although they may reduce the duration and severity of symptoms



Ascorbate



Monodehydroascorbate
(semidehydroascorbate)



Dehydroascorbate

Scurvy

- ▶ include skin changes, fragility of blood capillaries, gum decay, tooth loss, and bone fracture,
- ▶ many of which can be attributed to deficient collagen synthesis.

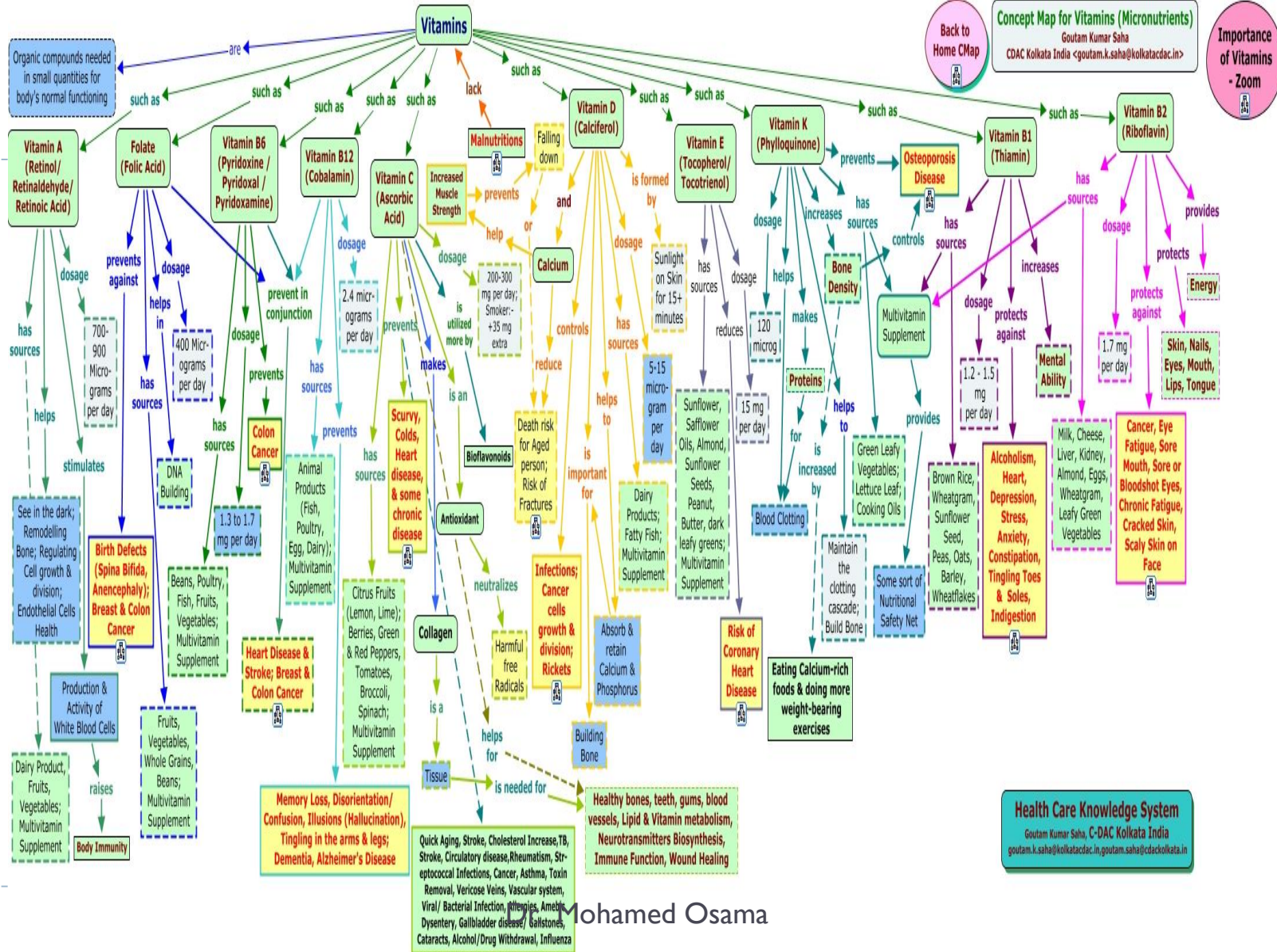
Scurvy

Ecchymoses



Bleeding gums-loose teeth





Fat Soluble Vitamins



Fat Soluble Vitamins

	Water-Soluble Vitamins: B Vitamins and Vitamin C	Fat-Soluble Vitamins: Vitamins A, D, E, and K
Absorption	Directly into the blood.	First into the lymph, then the blood.
Transport	Travel freely.	Many require protein carriers.
Storage	Circulate freely in water-filled parts of the body.	Stored in the cells associated with fat.
Excretion	Kidneys detect and remove excess in urine.	Less readily excreted; tend to remain in fat-storage sites.
Toxicity	Possible to reach toxic levels when consumed from supplements.	Likely to reach toxic levels when consumed from supplements.
Requirements	Needed in frequent doses (perhaps 1 to 3 days).	Needed in periodic doses (perhaps weeks or even months).

NOTE: Exceptions occur, but these differences between the water-soluble and fat-soluble vitamins are valid generalizations.

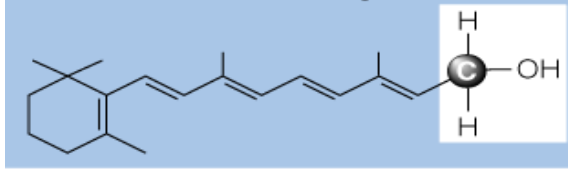
Fat Soluble Vitamins

- ▶ Are absorbed and transported by fat
- ▶ Excess vitamins are stored in the liver
- ▶ Body draws on stored vitamins when needed
- ▶ Large amounts can harm the body
- ▶ People who take vitamin supplements are advised to use caution

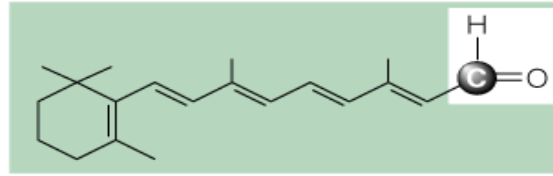
Vitamin A And Beta-Carotene

- ▶ Retinoids and carotenoids
- ▶ Vitamin A activity

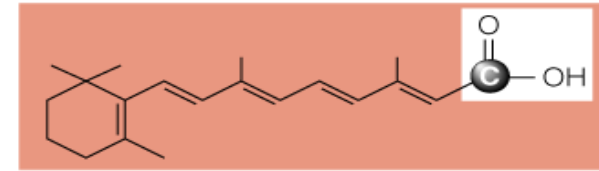
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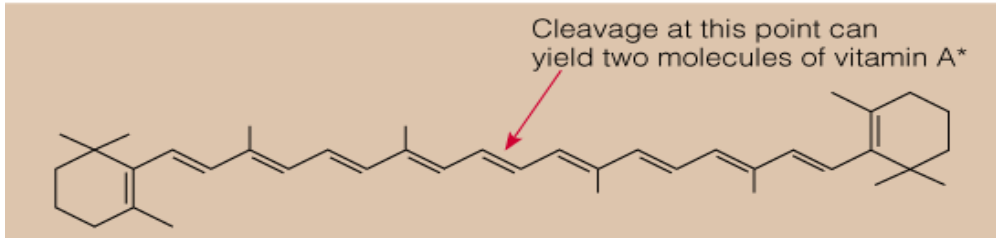
Retinol, the alcohol form



Retinal, the aldehyde form



Retinoic acid, the acid form



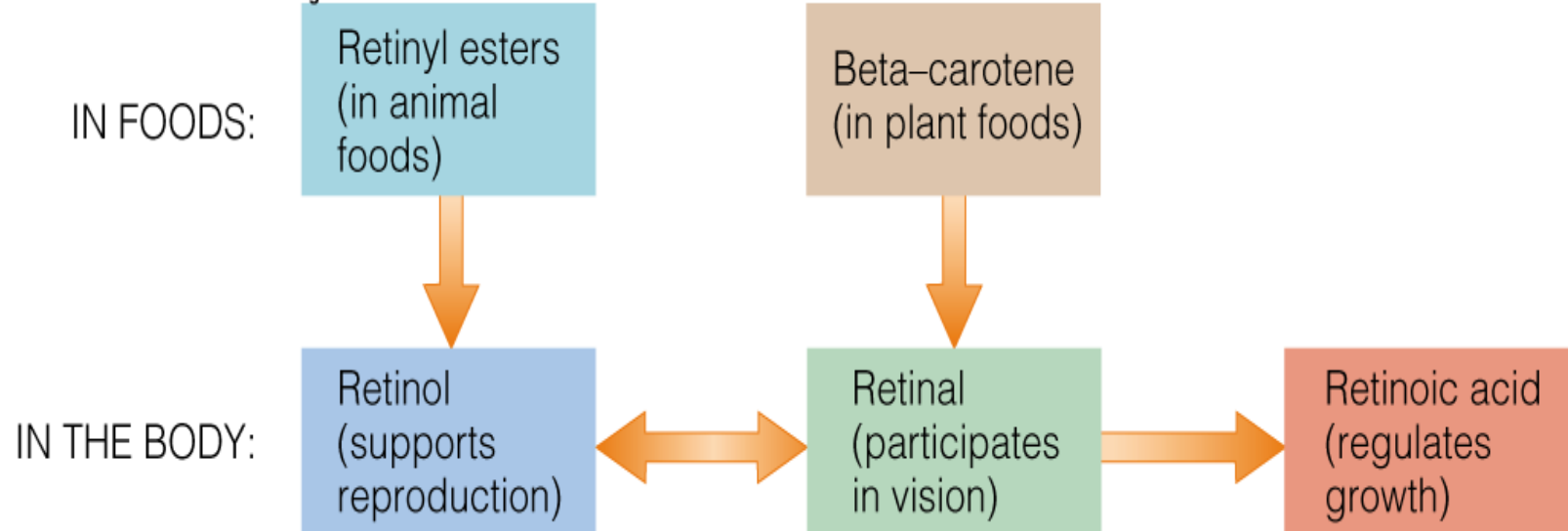
Beta-carotene, a precursor

Cleavage at this point can yield two molecules of vitamin A*

*Sometimes cleavage occurs at other points as well, so that one molecule of beta-carotene may yield only one molecule of vitamin A. Furthermore, not all beta-carotene is converted to vitamin A, and absorption of beta-carotene is not as efficient as that of vitamin A. For these reasons, 12 μ g of beta-carotene are equivalent to 1 μ g of vitamin A. Conversion of other carotenoids to vitamin A is even less efficient.

Vitamin A

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Significant sources

- ▶ Retinol: fortified milk, cheese, cream, butter, fortified margarine, eggs, liver
- ▶ Beta-carotene: dark leafy greens; broccoli, deep orange fruits (apricots, cantaloupe) and vegetables (carrots, sweet potatoes, pumpkin)

Role of Vitamin A:

- 1 . Vision
2. Vitamin A in protein synthesis and cell differentiation
- 3.Regulate gene expression

Vitamin A in the human body:

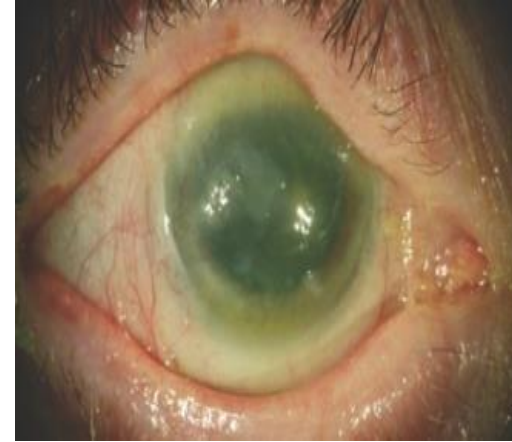
Other roles:

- ▶ Vitamin A in reproduction and growth
- ▶ Beta-carotene as an antioxidant

Vitamin A Deficiency:

- ▶ Hypovitaminosis A
- ▶ Symptoms:
 - ▶ Night blindness, corneal drying (xerosis), triangular gray spots on eye (Bitot's spots), softening of the cornea (keratomalacia), and corneal degeneration and blindness (xerophthalmia)

Deficiency and Toxicity

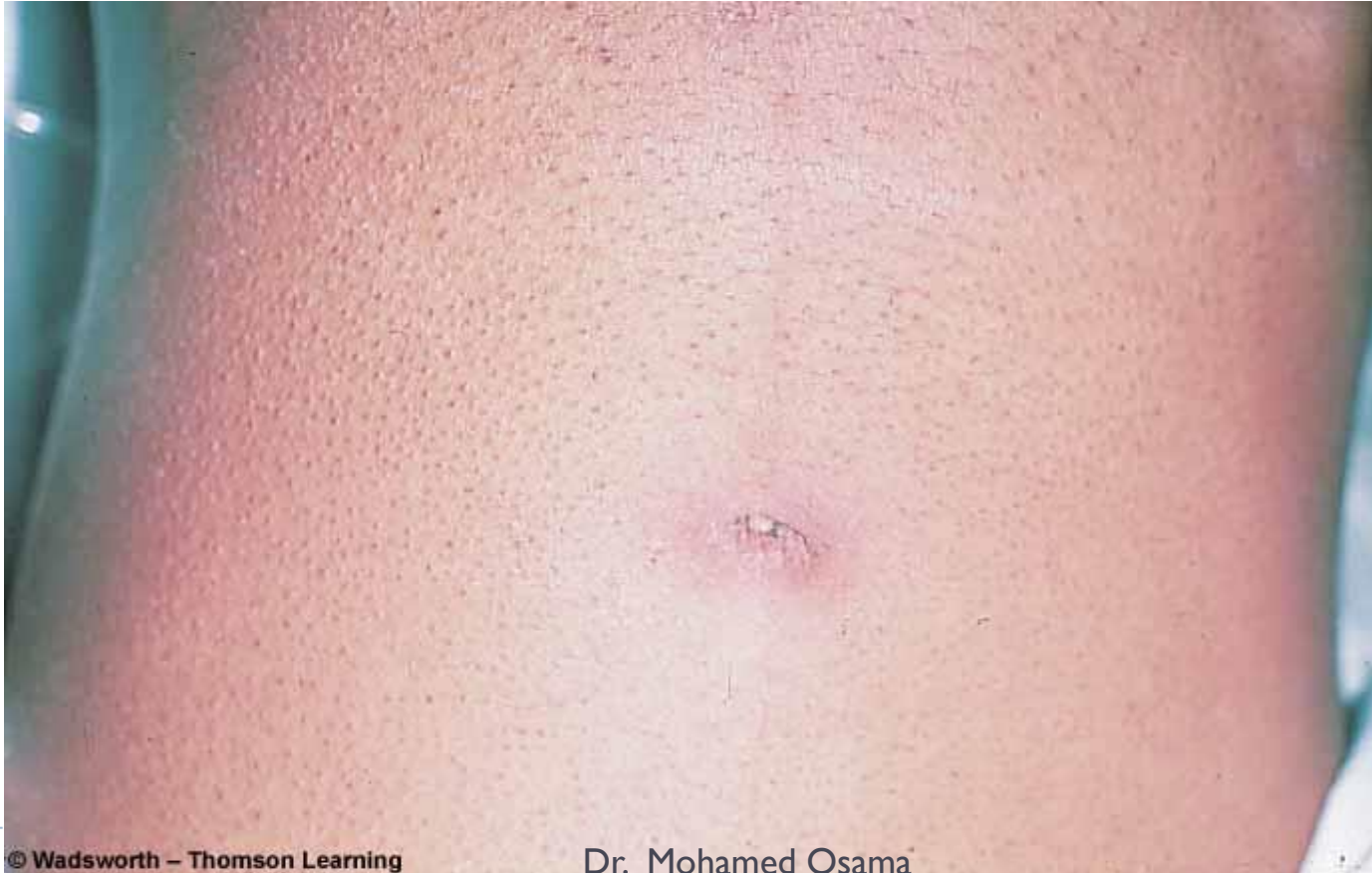


- ✓ Cell differentiation will diminish
- ✓ Keratin-producing cells replace mucus-secreting cells
- ✓ Night blindness, keratinization, xerophthalmia, impaired immunity, reproductive and growth abnormalities, exhaustion and death

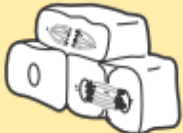
Vitamin A Deficiency

- ▶ Deficiency symptoms (continued)
 - ▶ Impaired immunity (infections)
 - ▶ Plugging of hair follicles with keratin, forming white lumps (hyperkeratosis)

Deficiency Symptom: Keratinization

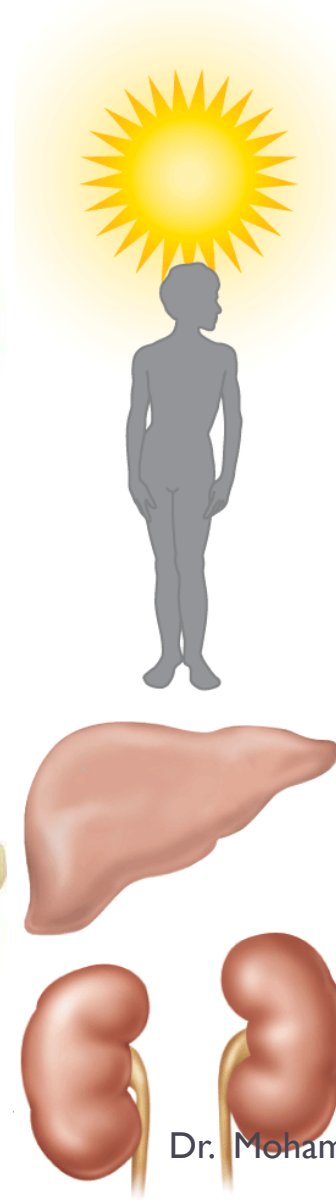


Deficiency and Toxicity

Deficient 0–500		Normal 500–3,000		Toxic 3,000 and over	
Effects on cells	Health consequences	Effects on cells	Health consequences	Effects on cells	Health consequences
Decreased cell division and deficient cell development 	Night blindness Keratinization Xerophthalmia Impaired immunity Reproductive and growth abnormalities Exhaustion Death	Normal cell division and development 	Normal body functioning	Overstimulated cell division 	Skin rashes Hair loss Hemorrhages Bone abnormalities Birth defects Fractures Liver failure Death

Vitamin D:

- ▶ Other names
 - ▶ Calciferol
 - ▶ 1,25-dihydroxy vitamin D (calcitriol)
 - ▶ Animal version: vitamin D₃ or cholecalciferol
 - ▶ Plant version: vitamin D₂ or ergocalciferol
 - ▶ Precursor is the body's own cholesterol



In the skin:

7-dehydrocholesterol
(a precursor made in the
liver from cholesterol)



Previtamin D₃



Vitamin D₃
(an inactive form)

In the liver:

Hydroxylation

25-hydroxy vitamin D₃

In the kidneys:

Hydroxylation

1,25-hydroxy vitamin D₃
(active form)

The sunshine vitamin: vitamin D.

Dr. Mohamed Osama

Role of Vitamin D:

- ▶ Chief functions in the body
 - ▶ Mineralization of bones (raises blood calcium and phosphorus by increasing absorption from digestive tract, withdrawing calcium from bones, stimulating retention by kidneys)

Vitamin D deficiency:

- ▶ Deficiency diseases:
 - ▶ Rickets
 - ▶ Osteomalacia



Bowed legs. In rickets, the poorly formed long bones of the legs bend outward as

weight-bearing activities such as walking begin.

Dr. Mohamed Osama

Vitamin D

- ▶ Inadequate calcification, resulting in misshapen bones (bowing of legs)
- ▶ Enlargement of ends of long bones (knees, wrists)
- ▶ Deformities of ribs (bowed, with beads or knobs)

Vitamin D

osteomalacia in adults

- ▶ Loss of calcium, resulting in soft, flexible, brittle, and deformed bones
- ▶ Progressive weakness
- ▶ Pain in pelvis, lower back, and legs

Vitamin E:

- ▶ Other name: alpha-tocopherol
- ▶ RDA:
 - ▶ Adults: 15 mg/day
- ▶ Upper level for adults:
 - ▶ 1000 mg/day

Sources:



- ▶ Significant sources
 - ▶ Polyunsaturated plant oils (margarine)
 - ▶ Leafy green vegetables, wheat germ, whole grains, liver, egg yolks, nuts, seeds

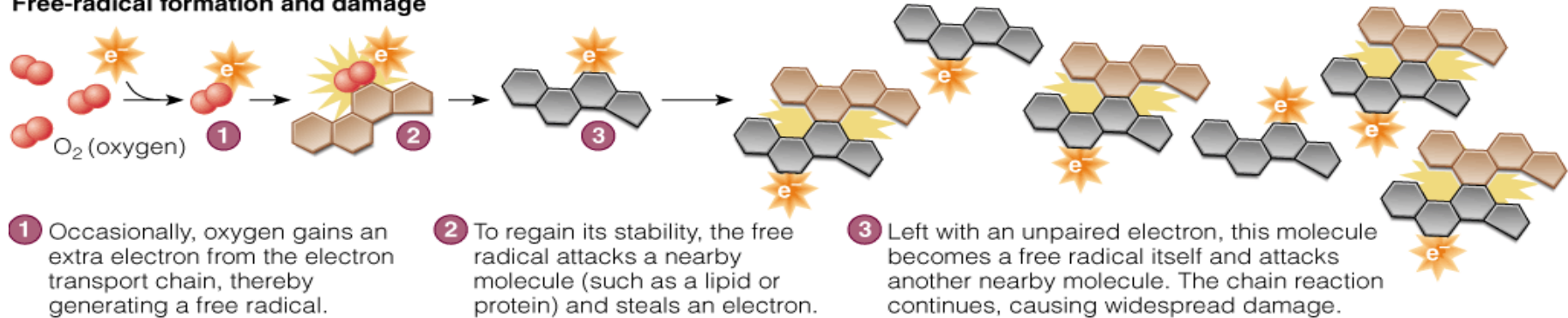
Vitamin E:

- ▶ Chief function in the body
 - ▶ Antioxidant:
stabilization of cell membranes, regulation of oxidation reactions, protection of polyunsaturated fatty acids and vitamin A)

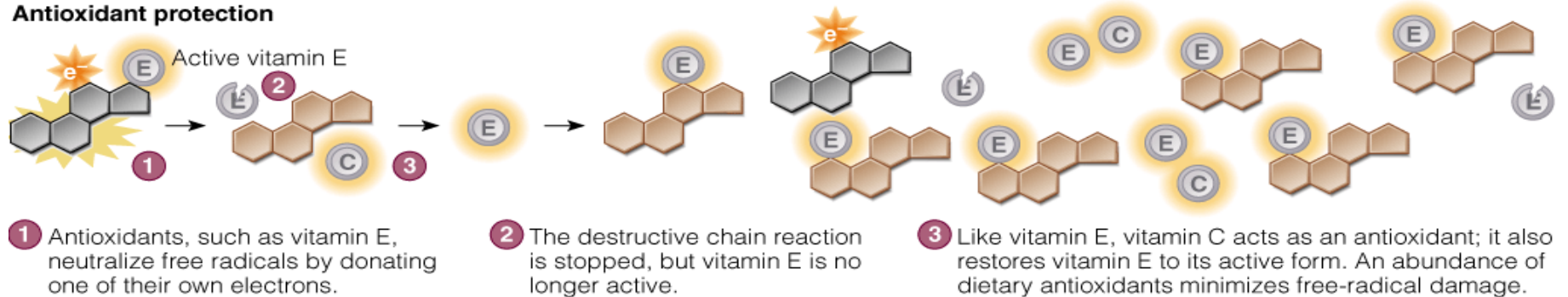
Vitamin E role:

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Free-radical formation and damage



Antioxidant protection



Deficiency and Toxicity:

- Deficiency symptoms
 - ▶ Red blood cell breakage
 - ▶ Nerve damage
- ▶ Toxicity symptoms
 - ▶ Augments the effects of anticoagulating medication

Vitamin K:

- ▶ **Other names**
 - ▶ Phylloquinone
 - ▶ Menaquinone
 - ▶ Menadione
 - ▶ Naphthoquinone
- ▶ **AI:**
 - ▶ Men: 120 µg/day
 - ▶ Women: 90 µg/day

sources:

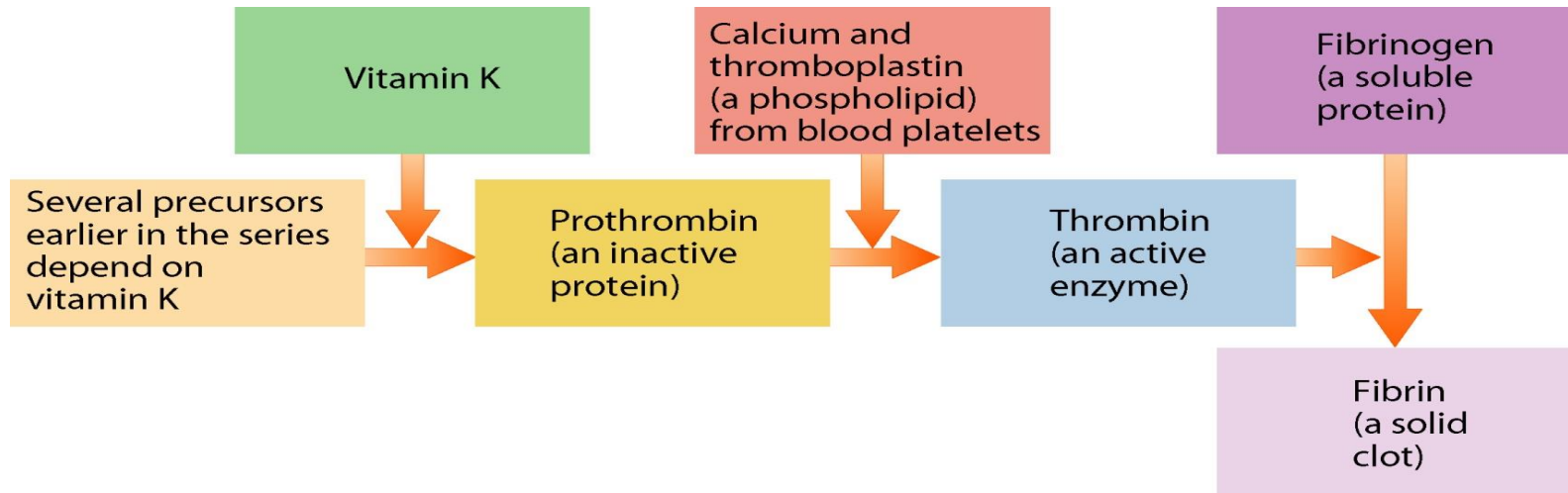
- ▶ Bacterial synthesis in the digestive tract
- ▶ Liver
- ▶ Leafy green vegetables, cabbage-type vegetables
- ▶ Milk



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Vitamin K Role:

- ▶ Chief functions in the body
 - ▶ Synthesis of blood-clotting proteins and bone proteins that regulate blood calcium



Vitamin K

- ▶ Deficiency symptoms
 - ▶ Hemorrhaging.

Deficiency:

- ▶ Vitamin K-deficiency may occur due to:
- ▶ Disturbed intestinal uptake (e.g: bile duct obstruction).
- ▶ Therapeutic or accidental intake of vitamin K-antagonists (e.g: warfarin therapy).
- ▶ Nutritional vitamin K-deficiency (very rare).

Thank you

